**Vijaya Ragunath S**
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## PROFESSIONAL SUMMARY

AI/ML Engineer specializing in LLMs, Retrieval-Augmented Generation (RAG), and end-to-end ML systems. Experienced in building scalable AI applications using FastAPI, Gemini API, and local LLMs (Ollama). Strong foundation in machine learning, deep learning, and data pipelines with hands-on experience delivering production-ready AI solutions with improved accuracy, reduced latency, and optimized inference costs. Passionate about designing intelligent systems that bridge real-world business problems with advanced AI.

## TECHNICAL SKILLS

- **Machine Learning:** Regression, Classification, Ensemble Learning (Random Forest, Gradient Boosting), Model Evaluation, Hyperparameter Tuning
- **Deep Learning:** ANN, CNN, RNN, LSTM
- **LLM & Generative AI:** RAG (Retrieval-Augmented Generation), Prompt Engineering, Gemini API, Ollama (Local LLM), Embeddings, Vector Databases
- **Frameworks & Libraries:** Scikit-learn, TensorFlow, Keras, Pandas, NumPy, Plotly
- **Backend & Deployment:** FastAPI, REST APIs, Model Deployment
- **Data Analysis & Visualization:** Power BI, Excel (Advanced), DAX
- **Tools & Platforms:** Git, Streamlit

## EXPERIENCE

**AI/ML Intern | VDart** **Nov 2025 – Mar 2026**

- Engineered and deployed end-to-end ML pipelines covering data preprocessing, feature engineering, model training, and evaluation, improving workflow efficiency by ~30% and enabling scalable model development.
- Developed a real-time LLM-powered chatbot using FastAPI and Gemini API, achieving low-latency responses (<1.5s) and seamless user interaction.
- Designed and implemented deep learning models (ANN, CNN, RNN) for both structured and unstructured data, enhancing predictive performance across multiple use cases.
- Integrated Ollama-based local LLM inference to enable hybrid AI architecture, reducing external API dependency and lowering inference costs by ~40%.
- Applied advanced prompt engineering techniques to optimize response quality, improving contextual accuracy and relevance by ~25%.
- Optimized API performance through efficient request handling and asynchronous processing, improving system scalability, reliability, and throughput.

**Data Analyst Intern | VDart** **May 2025 – Jun 2025**

- Developed interactive Power BI dashboards with drill-down capabilities and dynamic filtering, enabling data-driven decision-making and reducing manual reporting effort by ~25%.
- Cleaned, transformed, and validated large datasets using SQL queries and advanced Excel formulas, improving data accuracy and consistency by ~30%.
- Designed and implemented advanced DAX measures (CALCULATE, RANKX, ALL) to analyze key performance metrics and uncover actionable business insights.
- Automated reporting workflows and dashboard updates, improving analysis efficiency and reducing turnaround time for insights.
- Tools: Power BI, SQL, Excel, DAX

## PROJECTS

### AI Chatbot System (Company Project)

- Engineered a real-time LLM-powered chatbot using FastAPI, integrating Gemini API and Ollama for hybrid (cloud + local) inference.
- Architected a scalable API-driven system with advanced prompt engineering and response optimization, improving contextual accuracy and response relevance.
- Built and deployed an end-to-end AI application, seamlessly integrating backend services with frontend systems for real-world usability.
- Enabled low-latency, real-time conversational AI interactions, ensuring efficient request handling and smooth user experience.

### Document QA System (RAG) – Major Project

- Engineered an end-to-end Retrieval-Augmented Generation (RAG) system for question answering over unstructured documents, improving answer accuracy by ~35% compared to baseline LLM responses.
- Built a scalable document ingestion pipeline with parsing, intelligent chunking, and embedding generation, enabling efficient semantic retrieval across large document sets (1000+ documents).
- Designed and deployed a complete pipeline from document upload to answer generation using retrieval + LLM synthesis, reducing hallucinations by ~40% and improving response reliability.
- Optimized retrieval performance using embedding-based similarity search, reducing query response time to <2 seconds for real-time interaction.
- Enhanced contextual relevance through prompt engineering and retrieval tuning, significantly improving answer precision and user satisfaction.

## CERTIFICATIONS

- Python ML (LinkedIn)
- Applied ML (LinkedIn)
- Microsoft PL-300 (Power BI)

## EDUCATION

M.Sc. Applied Data Science | SRM Institute of Science and Technology |2024 – 2026
BSc, Computer Science | SRM Trichy Arts and Science College |2020 – 2023